

Lesson 03 Practice: Answers and Rationales

JKSSB Computer Awareness

October 6, 2026

PYQ provenance: Q1 is paraphrased from Website Operator 2026 Q72. The official provisional key is at [JKSSB](#); for Series A it identifies option B for Q72. The question wording here is not verbatim.

- Q1. B.** A bit is the smallest unit in the conventional hierarchy. A is not a basic unit in this list; C is eight bits; D is an architecture-dependent processor unit.
- Q2. B.** One nibble is four bits. A is two bits; C is one byte; D is larger than a nibble.
- Q3. C.** One byte contains eight bits. A is one nibble; B and D do not match the standard byte definition.
- Q4. A.** Two four-bit nibbles make one eight-bit byte. B has too few bits; C reverses the relationship; D would make 32 bits.
- Q5. B.** Lowercase **b** denotes bit. A byte uses uppercase B; C and D are not meanings of the standard rate symbol.
- Q6. C.** Mbps means megabits per second, a rate. MB and GB are storage amounts; bytes alone name an amount without a time denominator.
- Q7. C.** This lesson uses the traditional 1024 convention. A, B, and D are not equal to 1 KB under that convention.
- Q8. A.** $3 \times 1024 = 3072$ bytes. B uses 1000; C confuses bits with the conversion; D adds 3 to 1024 instead of multiplying.
- Q9. C.** $2 \times 1024 = 2048$ KB. A is half the correct value; B is the value for 1 MB; D confuses KB with bytes.
- Q10. A.** A byte contains eight bits, so $8/8 = 1$ MB/s. B omits the bit-to-byte conversion; C multiplies instead of dividing; D applies the inverse factor.
- Q11. B.** $64/8 = 8$ bytes. A is 32 bits; C is 128 bits; D confuses bits with bytes.
- Q12. C.** Word size depends on the processor architecture. A and B impose unsupported universal sizes; D wrongly equates a word with a character.
- Q13. D.** Word size is architecture-dependent, not universal. A, B, and C state the standard bit/byte conventions correctly.
- Q14. A.** MB denotes a byte-based storage amount. B would require /s; C changes the unit to megabits per second; D is unrelated to file capacity.

- Q15. D.** All three statements are true: 4 bits per nibble, 8 bits per byte, and two nibbles per byte. A, B, and C each omit a true statement.
- Q16. A.** $24/8 = 3$ MB/s before overhead. B skips conversion; C multiplies by eight; D divides by 80 rather than 8.
- Q17. A.** Capacity is how much data can be held; transfer rate is how much moves per time. B reverses the definitions; C confuses both with processor word size; D falsely assumes capacity determines speed.
- Q18. A.** Under the 1024 ladder, 1 GB=1024 MB. B is the decimal prefix relation; C is not a GB-to-MB conversion; D is a KB-to-byte relation.
- Q19. A.** Decimal prefixes use powers of 1000, while the traditional binary ladder uses powers of 1024. B, C, and D make unrelated claims about encoding, unit size, or network speed.
- Q20. B.** UTF-8 uses a variable number of bytes, so one byte per character is not guaranteed. A, C, and D are all consistent with variable-length encoding.